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Master's thesis

The title of my thesis

Any short subtitle

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0.1 Notation and important results

Start with some notation and conventions that will be used throughout the thesis.

We will now present some important results that will be used later in the thesis.

thm:bertini

Theorem 0.1.1 (Bertini). *If $\mathcal{D}$ is a linear system on a variety X in characteristic 0, then the general member of $\mathcal{D}$ is smooth away from the base locus of $\mathcal{D}$ and the singular locus of X .*

Add notation (this is from EH): A *linear system* on a scheme X is a pair $\mathcal{D} = (\mathcal{L}, V)$, where $\mathcal{L}$ is a line bundle on X and $V \subseteq H^0(X, \mathcal{L})$ a vector space of sections.

We refer to (!?) for a proof. ^{EH16} [EH16]

Find a good reference for this theorem.

thm:bezout

Theorem 0.1.2 (Bézout). *Let $X, Y \subseteq \mathbb{P}^n$ be two subvarieties having complementary dimension, i.e. $\dim(X) + \dim(Y) = n$. If X and Y intersect transversely, then they will intersect in exactly $\deg(X) \cdot \deg(Y)$ points.*

We refer to (!?) for a proof. ^{EH16} [EH16] corollary 2.4.

Find a good reference for this theorem. Don't require transverse intersections.

thm:fiber

Theorem 0.1.3 (Dimension formula for fibers). *Let k be a field and let $f: X \rightarrow Y$ be a dominant morphism of irreducible projective varieties over k . Let η denote the generic point of Y and write $X_\eta = X \times_Y \operatorname{Spec} k(Y)$ for the generic fibre. Then*

$$\dim X = \dim Y + \dim X_\eta.$$

Write the proof, and/or find reference. Maybe use Vakil 12.4.2.

0.2 Blow-ups, and the exceptional divisor

Decide if this is viable/necessary for the thesis. Technically use this in the 27 lines proof, but could assume this is known, and just give blow ups in relation to divisors.

0.3 Divisors

Decide if this is viable/necessary for the thesis. We do use this in the 27 lines proof, specifically self-intersection. This could be assumed known. Could stick to Beauville.

Contents

Chapter 1

Grassmannians and the space of lines in $\mathbb{P}^3$

Grassmannians

In this chapter we will introduce Grassmann varieties, or Grassmannians, and focus on $\mathbb{G}(1, 3)$, which is the space of lines in $\mathbb{P}^3$. We will also introduce the Plücker embedding, which is a way to embed the Grassmannian into a projective space, giving it the structure of a projective variety. For a more detailed introduction to Grassmannians, we refer to [EH16].

1.1 Definition and basic properties

Remark 1.1.1. We will use k to refer to the dimension of a linear subspace, and $\mathbb{k}$ to refer to a field.

A Grassmannian is a projective variety whose closed points correspond to linear subspaces of a certain dimension in an n -dimensional vector space. As a set, $\text{Gr}(k, V)$ is the set of all k -dimensional linear subspaces of the vector space V . In Section 1.2, we will give it the structure of a projective variety. In our case, V will be a vector space over $\mathbb{k}$.

$$\text{Gr}(k, n)(\mathbb{k}) = \{[L] \mid L \subseteq \mathbb{k}^n \text{ is a linear space of dimension } k\}.$$

A k -dimensional linear subspace of an n -dimensional vector space V can also be thought of as a $(k - 1)$ -dimensional linear subspace of $\mathbb{P}V \cong \mathbb{P}^{n-1}$. Due to this, we will often write $\text{Gr}(k, V)$ as $\mathbb{G}(k - 1, \mathbb{P}V)$. When there is no need to specify the vector space V , only its dimension n , we will write $\text{Gr}(k, n) = \mathbb{G}(k - 1, n - 1)$.

[EO25] A dimension $n - k$ linear subspace $L \subseteq \mathbb{k}^n$ is described as the kernel of a surjective linear map $\mathbb{k}^n \rightarrow \mathbb{k}^{n-k}$, represented by an $(n - k) \times n$ matrix a of rank $n - k$:

$$L = \text{Ker}(A) \subseteq \mathbb{k}^n,$$

where

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{(n-k),1} & a_{(n-k),2} & \cdots & a_{(n-k),n} \end{pmatrix}.$$

Note that the matrix A is not unique for a given linear subspace L , since multiplying A on the left by an invertible $(n - k) \times (n - k)$ matrix G gives another matrix GA with

the same kernel as A . Thus, we consider the equivalence relation on the set of $(n-k) \times n$ matrices of rank $n-k$ given by $A \sim A'$ if there exists an invertible $(n-k) \times (n-k)$ matrix G such that $A' = GA$. The Grassmannian $\text{Gr}(k, n)$ can then be identified with the set of equivalence classes of $(n-k) \times n$ matrices of rank $n-k$ under this equivalence relation.

That is, we want to consider the set

$$(\mathbb{A}^{(n-k)n}(\mathbb{k}) - Z)/\text{GL}_{n-k}(\mathbb{k})$$

^[E025]
[EO25] Where Z is the set of $(n-k) \times n$ matrices of rank less than $n-k$, and $\mathbb{A}^{(n-k)n}(\mathbb{k})$ parametrizes the set of all $(n-k) \times n$ matrices with entries in $\mathbb{k}$.

1.2 The Plücker embedding

sec:Plucker

- General idea of Plucker embedding, referencing EO and EH.
- Example: $\mathbb{G}(1, 3)$.

1.3 Covering by affine spaces

- Idea of covering by affine spaces, referencing EO and EH.
- Example: $\mathbb{G}(1, 3)$.

Example 1.3.1. For $n = 4, k = 2$, there are 6 affine spaces $\mathbb{A}^4$. They are:

$$U_{1,2} = \begin{pmatrix} 1 & 0 & a_{1,3} & a_{1,4} \\ 0 & 1 & a_{2,3} & a_{2,4} \end{pmatrix}$$

$$U_{1,3} = \begin{pmatrix} 1 & a_{1,2} & 0 & a_{1,4} \\ 0 & a_{2,2} & 1 & a_{2,4} \end{pmatrix}$$

$$U_{1,4} = \begin{pmatrix} 1 & a_{1,2} & a_{1,3} & 0 \\ 0 & a_{2,2} & a_{2,3} & 1 \end{pmatrix}$$

$$U_{2,3} = \begin{pmatrix} a_{1,1} & 1 & 0 & a_{1,4} \\ a_{2,1} & 0 & 1 & a_{2,4} \end{pmatrix}$$

$$U_{2,4} = \begin{pmatrix} a_{1,1} & 1 & a_{1,3} & 0 \\ a_{2,1} & 0 & a_{2,3} & 1 \end{pmatrix}$$

▲

1.4 Universal sub and quotient bundles

Fill in. Include prop 3.3 from EH16.

^[EH16]
[EH16] Let V be an n -dimensional vector space over a field $\mathbb{k}$. Consider the Grassmannian $\text{Gr}(k, V)$ of k -dimensional linear subspaces of V . Let $\mathcal{V} = \text{Gr}(k, V) \times V$ be the trivial vector bundle over $\text{Gr}(k, V)$ with fiber V .

Definition 1.4.1. The *universal subbundle* $\mathcal{S}$ over $\mathrm{Gr}(k, V)$ is the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$. That is,

$$\mathcal{S}_{[L]} = L \subseteq V = \mathcal{V}_{[L]}.$$

The quotient $\mathcal{Q} = \mathcal{V}/\mathcal{S}$ is called the *universal quotient bundle* over $\mathrm{Gr}(k, V)$.

We define the *universal subbundle* $\mathcal{S}$ over $\mathrm{Gr}(k, V)$ as the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$. Similarly, we define the *universal quotient bundle* $\mathcal{Q}$ over $\mathrm{Gr}(k, V)$ as the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the quotient space V/L .

Proposition 1.4.2 ([EH16]^{EH16}, Proposition 3.3). *The subset $\mathcal{S}$ of $\mathcal{V}$ whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$ is a vector bundle over $\mathrm{Gr}(k, V)$.*

1.5 Short on Schubert cycles (?)

Decide if we should include or not. Probably not.

1.6 Incidence correspondence and the universal Fano scheme

- Introduce the universal Fano scheme, and incidence correspondences.
- State the dimension of the universal Fano scheme, and the expected dimension of the Fano scheme of a general hypersurface.
- Explain how this tells us whether to expect a hypersurface to contain lines, and of which dimension those lines are.
- State ^{cor:Fano}1.6.6, which tells us when to expect lines on a general surface in $\mathbb{P}^3$.

We aim to introduce the powerful concept of incidence correspondences (or an incidence variety), which will be a tool we will use several times later in this thesis. We do this by introducing the universal Fano scheme, which is an example of an incidence variety. Generally, an incidence correspondence is a set of pairs (x, y) such that x lies in y , where x and y are elements of X and Y respectively, for some varieties(!?) X and Y . This can be visualized in the following diagram, with projection maps π_1 and π_2 :

$$\begin{array}{ccc} & \mathcal{I} = \{(x, y) \in X \times Y \mid x \text{ lies in } y\} & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ X & & Y \end{array} \quad (1.1)$$

Now we define an incidence correspondence of k -planes $L \in \mathbb{G}(k, n)$ contained in degree d surfaces in $\mathbb{P}^n$.

Definition 1.6.1. (^{EH16}[EH16] Section 6.1). Let $N = \binom{n+d}{d} - 1$, so that $\mathbb{P}^N$ parametrizes all degree d hypersurfaces in $\mathbb{P}^n$. The *universal Fano scheme* is defined as

$$\Phi(n, d, k) = \{(X, L) \in \mathbb{P}^N \times \mathbb{G}(k, n) \mid L \subseteq X\}$$

ark:universal Fano

Remark 1.6.2. The *Fano scheme* $F_k(X) \subseteq \mathbb{G}(k, n)$ of a variety $X \subseteq \mathbb{P}^n$ parametrizes k -planes on X . The universal Fano scheme parametrizes k -planes for all varieties of a certain degree in $\mathbb{P}^n$. By taking the fiber over a point $[X] \in \mathbb{P}^N$, that being the class of the surface $X \subseteq \mathbb{P}^n$, we get the Fano scheme $F_k(X)$ for the specified variety. In this sense, taking the fiber over a "general" X results in a "general" Fano scheme $F_k(X)$. Studying the universal Fano scheme can then tell us what we can "expect" when examining a specific Fano scheme. It is therefore a very useful tool in telling us about a general member of the varieties.

Lemma 1.6.3. $\Phi \subset \mathbb{P}^N \times \mathbb{G}(k, n)$ is a closed subset of $\mathbb{P}^N \times \mathbb{G}(k, n)$.

Decide if we want to prove, or refer to proof. Can refer to Harris 1995 chapter 6, or perhaps proof of 24.31 in EO

Proposition 1.6.4 (^{EH16}[EH16], Proposition 6.1). Let $N = \binom{n+d}{d} - 1$. The universal Fano scheme $\Phi = \Phi(n, d, k) \subseteq \mathbb{P}^N \times \mathbb{G}(k, n)$ is a smooth irreducible variety of dimension

$$\dim \Phi(n, d, k) = N + (k+1)(n-k) - \binom{k+d}{k}.$$

We ask when we expect a general smooth surface X of degree d to contain lines, and remember that this question is answered by looking at the geometry of the Fano scheme $F_k(X)$ for a general smooth surface X . More precisely one asks what dimension the Fano scheme has, therefore deciding what dimension the set of lines in our general surface is. As noted in Remark 1.6.2, $F_k(X)$ can be obtained by taking the fiber over a point $[X] \in \mathbb{P}^N$. We therefore ask what dimension this fiber will have. From Theorem 0.1.3 we get that $\dim \pi^{-1}([X]) = \dim \Phi - \dim \mathbb{P}^N$. We define this as $\phi(n, d, k) := \dim \Phi - N$. Now we can answer the question of when we expect lines to be contained in our surface by looking at $\phi(n, d, k)$:

Corollary 1.6.5 (^{EH16}[EH16], corollary 6.2). *(a) The dimension of any component of the family of k -planes on any hypersurface of degree d in $\mathbb{P}^n$ is at least*

$$\phi(n, d, k) := (k+1)(n-k) - \binom{k+d}{k}.$$

- (b) If $\phi(n, d, k) < 0$, then the general hypersurface of degree d in $\mathbb{P}^n$ contains no k -planes.*
- (c) If $\phi(n, d, k) \geq 0$, and the general hypersurface of degree d contains any k -planes, then every hypersurface of degree d contains k -planes, and every component of the family of k -planes on a general hypersurface of degree d has dimension exactly $\phi(n, d, k)$.*

Proof.

Decide if we want to prove, or refer to proof.

□

To bring this back to a familiar context of lines on surfaces in $\mathbb{P}^3$, we set $n = 3$ and $k = 1$. We then get that $\phi(3, d, 1) = 3 - d$. We use this in the following corollary:

cor:Fano

Corollary 1.6.6. (^{E025}[EO25], 22.31) *For a hypersurface $X \subseteq \mathbb{P}^3$ of degree d , we expect to find no lines on X when $d \geq 4$, finitely many lines when $d = 3$, and positive-dimensional families of lines when $d \leq 2$.*

1.7 The Chow ring

^[FUL84]
[Ful84] Let X be an algebraic scheme. A k -cycle on X is a finite formal sum

$$\sum_i m_i V_i$$

where the V_i are k -dimensional subvarieties of X , and the m_i are integers. The group of k -cycles on X , denoted $Z_k(X)$, is the free Abelian group generated by the k -cycles.

Example 1.7.1. $(n - 1)$ -cycles are Weil divisors. ▲

1.8 Chern classes

Briefly introduce Chern classes. Include Whitney sum formula and the splitting principle.

1.8.1 The first Chern class of a line bundle

^[EH16]
[EH16], 1.4.

Let $\mathcal{L}$ be a line bundle on a variety X and σ be a rational section. Then on an open affine set U of a covering of X , we can write $\sigma|_U = f_U g_U$, and define $\text{Div}(\sigma)|_U = \text{Div}(f_U) - \text{Div}(g_U)$. This definition agrees on the intersections of the open affine sets, and thus defines a divisor on X , which is a *Cartier divisor*.

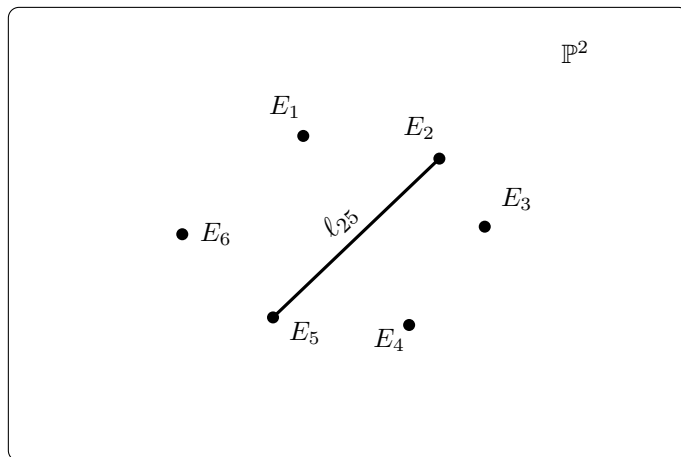


Figure 1.1: The projective plane $\mathbb{P}^2$ with six marked blow-up points and a strict transform of the line ℓ_{25} .

fig:blowup6

1.9 Lines on a cubic surface

From Corollary 1.6.6, we know that a smooth cubic surface contains finitely many lines. We will now show that it contains exactly 27 lines. This is a very famous result, and it was first proved in 1849 by Arthur Cayley and George Salmon, in a proof we will give a sketch of(!?).

Consider continuing this historical view.

Theorem 1.9.1. *Each smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines.*

This will be the main theorem of this section. We will prove it in two different ways. The first will be a well known proof, relating cubic surfaces to blow-ups of $\mathbb{P}^2$ in 6 points. The second proof will use intersection theory on the Grassmannian $\mathbb{G}(1, 3)$.

Describe second result better.

We begin with the first proof.

1.9.1 Cubic surfaces as blow-ups of $\mathbb{P}^2$ in 6 points

Make a better figure of the blow-up of $\mathbb{P}^2$ in 6 points.

This proof will rely on theory which is not in the scope of this thesis, but it will allow us to draw a picture to show a representation of the 27 lines themselves, which we will use later.

The points $P_1, \dots, P_6$ in $\mathbb{P}^2$ are said to be in *general position* if no three of the points are collinear, and no six lie on a conic.

prop:blowup6

Proposition 1.9.2. (^{Ha77}[Har77], 4.7/4.7.3) *Let $P_1, \dots, P_6 \in \mathbb{P}^2$ be in general position. Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X is isomorphic to the blow-up of $\mathbb{P}^2$ in the points $P_1, \dots, P_6$.*

This is part of a bigger class of surfaces, those being the *Del Pezzo surfaces*. These are surfaces which (Consider continuing, and maybe cite Beauville if so)(!?)

prop:27

Proposition 1.9.3. (^{Ha77}[Har77], 4.9) *Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X contains exactly 27 distinct lines. Each having self-intersection -1 , and they are the only irreducible curves with negative self-intersection. They are:*

- *The exceptional curves $E_1, \dots, E_6$ (6 of these)*
- *The strict transform of lines in $\mathbb{P}^2$ containing pairs of points P_i, P_j , $i \neq j$ (15 of these).*
- *The strict transform of conics in $\mathbb{P}^2$ not containing P_i (6 of these)*

1.9.2 Cayley-Salmon (!?)

Decide if we want to include this classic proof. We could do as EO, Finding 27 lines on the Fermat cubic.

1.9.3 Proof by degree of Chern classes

Do proof in EH16.

1.10 Eckardt points

Introduce Eckardt points, how many there are in cubic surface in $\mathbb{P}^3$, and the pencil example. Decide if this is viable for the thesis.

Chapter 2

Results

results

2.1 Surfaces including intersecting lines

Align name of lines better

In this section we will study the problem of finding a smooth hypersurface $X_d \subset \mathbb{P}^3$ of degree d , containing a k -cross.

Definition 2.1.1. A k -cross is a union of k lines $l_1, l_2, \dots, l_k$ in $\mathbb{P}^3$ which intersect at a common point p and lying in a common plane H . We will write $Z_k = \bigcup_{i=1}^k l_i$.

We require the lines to lie in a plane because if this is not true, then the point p gives rise to a tangent space of dimension 3, which cannot be included in a surface in $\mathbb{P}^3$.

Make this argument rigorous, maybe a proposition.

It is worth noting the specific question we are asking, which is that given a k -cross, does there exist a smooth hypersurface of degree d including it. We do not ask whether a k -cross is to be expected inside a general hypersurface.

Remark 2.1.2. Note that for automorphisms of $\mathbb{P}^3$, lines are sent to lines. Therefore, we can study the lines $l_1 = (\alpha : \beta : 0 : 0)$, $l_2 = (\alpha : 0 : \gamma : 0)$, $l_3 = (\alpha : \delta : \lambda_3 \delta : 0)$, $\dots$, $l_k = (\alpha : \delta : \lambda_k \delta : 0)$ for $\lambda_i \in \mathbb{N}$ by taking the automorphism of $\mathbb{P}^3$ which sends these lines to the given lines, and then study the hypersurface of degree d in $\mathbb{P}^3$ which contains these lines. This automorphism will also take a smooth hypersurface of degree d to a smooth

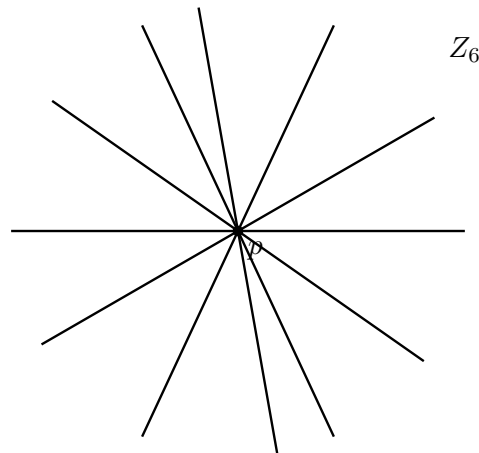


Figure 2.1: An example of a 6-cross in $\mathbb{P}^3$.

fig:Z_6

hypersurface of degree d , so showing properties for lines in this form is sufficient for the general case.

We are interested in both the existence and non-existence of degree d hypersurfaces containing k -crosses. We first consider surfaces with degree less than k , but more than 1 (degree 1 provides us a simple choice of surface including a k -cross, namely H).

Lemma 2.1.3. *Given a k -cross, there does not exist any smooth hypersurface of degree $1 < d < k$ such that the lines are contained in the hypersurface.*

Proof. Assume for contradiction that there exists a smooth hypersurface of degree $1 < d < k$ such that the lines are contained in the hypersurface. Then we can fix another line ℓ in H , which does not intersect p , and which does not lie in the hypersurface X_d . ℓ will intersect each of the lines at a point p_i , all being distinct since the lines only intersect in p . ℓ will also intersect X_d at the points p_i , thus intersecting the hypersurface at k points. By Bezout's theorem (Theorem 0.1.2), the intersection of a line with a hypersurface of degree d is at most d points, giving a contradiction.

Can't the line stay on our surface when going from one line to the other?

□

We now move on to the case of existence, with the following proposition:

Proposition 2.1.4. *For a given k -cross, we can find a smooth hypersurface of degree k such that the lines are contained in the hypersurface.*

Proof. Let $\mathfrak{L}$ be the linear system of hypersurfaces of degree k in $\mathbb{P}^3$ containing the lines $l_0, l_1, \dots, l_k$ (Prove this is not empty, because of singular surfaces.)(!). Recall that Bertini's theorem (Theorem 0.1.1) tells us that the general member of $\mathfrak{L}$ is smooth outside the base locus of $\mathfrak{L}$. The base locus of $\mathfrak{L}$ is the union of the lines $l_0, l_1, \dots, l_k$. Thus, we must find a function $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_k$ such that $V(f)$ is smooth on the lines $l_0, l_1, \dots, l_k$. Bertini's theorem then gives us that we can find a hypersurface which is also smooth outside the base locus of $\mathfrak{L}$.

We study the function $f = g(x_1, x_2) + h(x_0, x_3)$, where $h = x_0^{k-1}x_3$ and $g = \prod_{i=3}^k l_i = x_1x_2 \prod_{i=3}^k (x_1 - \lambda_i x_2)$, giving the following polynomial:

$$f(x_0, x_1, x_2, x_3) = x_1x_2 \prod_{i=3}^k (x_1 - \lambda_i x_2) + x_0^{k-1}x_3$$

We can see that f is a polynomial of degree k , and that f includes the lines $l_0, l_1, \dots, l_k$, since $f(\alpha : \beta : 0 : 0) = f(\alpha : 0 : \gamma : 0) = f(\alpha : \delta : \lambda_i \delta : 0) = 0$. We must then check that f is smooth on the base locus of $\mathfrak{L}$, which is to check that $\bigcap_{i=0}^3 (\partial_i f = 0) = \emptyset$.

We get the following derivatives:

$$\begin{aligned} \partial_0 f &= \partial_0 g + \partial_0 h = 0 + (k-1)x_0^{k-2}x_3 = (k-1)x_0^{k-2}x_3 \\ \partial_1 f &= \partial_1 g + \partial_1 h = x_2 \prod_{i=3}^k (x_1 - \lambda_i x_2) + x_1x_2 \sum_{i=3}^k \prod_{\substack{j=3 \\ j \neq i}}^k (x_1 - \lambda_j x_2) \\ \partial_2 f &= \partial_2 g + \partial_2 h = x_1 \prod_{i=3}^k (x_1 - \lambda_i x_2) + x_1x_2 \sum_{i=3}^k (-\lambda_i) \cdot \prod_{\substack{j=3 \\ j \neq i}}^k (x_1 - \lambda_j x_2) \end{aligned}$$

$$\partial_3 f = \partial_3 g + \partial_3 h = 0 + x_0^{k-1} = x_0^{k-1}$$

We then find that

- $\partial_0 f = 0$ on the entire base locus.
- $\partial_1 f = 0$ on the line $l_0 = (\alpha : \beta : 0 : 0)$.
- $\partial_2 f = 0$ on the line $l_1 = (\alpha : 0 : \gamma : 0)$.
- $\partial_3 f = 0$ if and only if $x_0 = 0$, which is the case for the following points: $(0 : \beta : 0 : 0), (0 : 0 : \gamma : 0), (0 : \delta : \lambda_i \delta : 0)$ for $3 \leq i \leq k$.

We can see that the intersection of all of these zero loci is empty, showing that the function f is smooth on the base locus of $\mathfrak{L}$. Thus, we can find a smooth hypersurface of degree k such that the lines are contained in the hypersurface. □

By the results above, we get the following corollary:

Corollary 2.1.5. *Given a k -cross, we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface if and only if $d \geq k$ or $d = 1$.*

Proof. If we have k lines, we can find a smooth hypersurface of degree $d \geq k + 1$ by including arbitrary lines $l_{k+1}, l_{k+2}, \dots, l_d$ lying in the plane. For the case $d = 1$, we can take the plane H containing the lines. □

2.2 The set of degree d surfaces including k -crosses

In the previous section we studied the existence and non-existence of smooth hypersurfaces of degree d including a k -cross. We will now study the dimension of the set of degree d surfaces including k -crosses. Let Z_k be a k -cross lying in a plane H , which we will assume is the plane $V(x_3)$ by an automorphism of $\mathbb{P}^3$.

Proposition 2.2.1. *Let $A_k = \{S_d \subseteq \mathbb{P}^3 \mid Z_k \subset S_d\} \subseteq \{S_d \subseteq \mathbb{P}^3\}$ be the set of degree d surfaces containing a general Z_k . Then $\dim A_k = \binom{d-k+2}{2} + \binom{d+2}{3} - 1$.*

Proof. Let $S_d = V(f)$ for some $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_d$ be a surface which includes Z_k . Since Z_k is included in H , we write f as $f = g(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$, separating the part of f , $x_3 h(x_0, x_1, x_2, x_3)$, which vanishes on the plane. Saying that $Z_k \subseteq S_d$ is the same as saying $f|_{Z_k} = 0$, which implies that $g(x_0, x_1, x_2)|_{Z_k} = 0$, since $x_3 h(x_0, x_1, x_2, x_3)|_{Z_k}$ already vanishes. This implies that $l_1 l_2 \cdots l_k$ divides $g(x_0, x_1, x_2)$, meaning $g(x_0, x_1, x_2) = l_1 l_2 \cdots l_k \cdot j(x_0, x_1, x_2)$ for some $j(x_0, x_1, x_2) \in \mathbb{C}[x_0, x_1, x_2]_{d-k}$. Thus, we can write f as $f = l_1 l_2 \cdots l_k \cdot j(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$, where $h(x_0, x_1, x_2, x_3) \in \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1}$.

Since we can choose $j(x_0, x_1, x_2)$ and $h(x_0, x_1, x_2, x_3)$ freely, we get that $\dim A_k = \dim \mathbb{C}[x_0, x_1, x_2]_{d-k} + \dim \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1} = \binom{d-k+2}{2} + \binom{d+2}{3} - 1$.

Should we subtract 1 to account for it being a projective space?

□

Include examples?

Continue with the incidence correspondence?

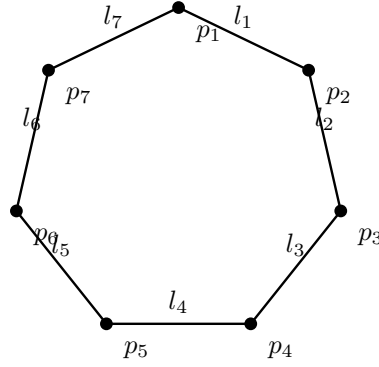


Figure 2.2: 7-gon drawn in a plane; each vertex is an intersection point p_i and the line segments represent the lines l_i .

fig:k-gon

2.3 Surfaces including k-gons

In this section, we will consider k -gons in $\mathbb{P}^3$ and their relationship with hypersurfaces.

Definition 2.3.1. A k -gon is a union of k lines $l_1, l_2, \dots, l_k$ such that l_i intersects l_{i+1} for $1 \leq i < k$, and l_k intersects l_1 . We will write $C_k = \bigcup_{i=1}^k l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \leq i < k$, and $p_k = l_k \cap l_1$.

Make lines go past intersection points.

prop:k-gon

Proposition 2.3.2. Let C_k be a k -gon in $\mathbb{P}^3$. Then we can find a hypersurface of degree d such that $C_k \subseteq S_d$ when $k \leq \frac{(d+3)(d+2)(d+1)}{6d}$.

Proof. We can look at the zero cohomology of the ideal sheaf $\mathcal{I}_{C_k}$ of the k -gon to see if the k -gon can be contained in a hypersurface. We have the short exact sequence

Why?

$$0 \rightarrow \mathcal{I}_{C_k} \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_{C_k} \rightarrow 0.$$

We twist this by $\mathcal{O}_{\mathbb{P}^3}(d)$ to get

$$0 \rightarrow \mathcal{I}_{C_k}(d) \rightarrow \mathcal{O}_{\mathbb{P}^3}(d) \rightarrow \mathcal{O}_{C_k}(d) \rightarrow 0.$$

From this we get the long exact sequence in cohomology:

$$\begin{array}{ccccccc} 0 & \rightarrow & H^0(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) & \rightarrow & H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) & \rightarrow & H^0(C_k, \mathcal{O}_{C_k}(d)) \\ & & \searrow & & \searrow & & \searrow \\ & & H^1(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) & \rightarrow & H^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) & \rightarrow & H^1(C_k, \mathcal{O}_{C_k}(d)) \\ & & \searrow & & \searrow & & \searrow \\ & & H^2(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) & \rightarrow & H^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) & \rightarrow & \dots \end{array}$$

We know that $H^i(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = 0$ for $i > 0$ (cite)(!?). This means we can focus on the first part of the sequence:

$$0 \rightarrow H^0(\mathbb{P}^3, \mathcal{I}_{C_k}) \rightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) \rightarrow H^0(C_k, \mathcal{O}_{C_k}) \rightarrow H^1(\mathbb{P}^3, \mathcal{I}_{C_k}) \rightarrow 0.$$

Since taking dimensions in cohomology is additive, we get

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_k}) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) + h^1(\mathbb{P}^3, \mathcal{I}_{C_k}) - h^0(C_k, \mathcal{O}_{C_k}).$$

By Thm 18.4 (!?) in ^{E025}[EO25] we get that $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = \binom{d+3}{3}$. $h^0(C_k, \mathcal{O}_{C_k}(d))$ can be computed by considering the short exact sequence:

$$0 \rightarrow \mathcal{O}_{C_k}(d) \rightarrow \bigoplus_{i=1}^k \mathcal{O}_{l_i}(d) \rightarrow \bigoplus_{i=1}^n \mathcal{O}_{p_i} \rightarrow 0.$$

This gives us $h^0(C_k, \mathcal{O}_{C_k}) = k \cdot \binom{d+1}{1} - k = kd$.

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_k}) \geq \binom{d+3}{3} - kd.$$

Meaning we can guarantee a degree d surface when $\binom{d+3}{3} - kd \geq 0$, or equivalently when $k \leq \frac{(d+3)(d+2)(d+1)}{6d}$. □

Make proof more coherent, we jump around a bit.

To get an exact value for $h^0(\mathbb{P}^3, \mathcal{I}_{C_k})$, we would need to compute $h^1(\mathbb{P}^3, \mathcal{I}_{C_k})$. This issue will be explored further in Section 2.5.

Explain significance of h^0 , and do it before this proof.

Remark 2.3.3. Note that this argument does not guarantee that the surface is smooth, only that it exists.

2.4 Computational results

We should probably move the code to the appendix, and just discuss the results here.

We have done computational experiments in Macaulay2 to get a better understanding of the inequality in Proposition 2.3.2. We have tested the existence and smoothness of hypersurfaces including k -gons for various degrees d and number of lines k . The code used for these experiments is included in the following snippet. We will discuss the details of the code below, along with its results.

lst:k-gon

Listing 2.1: k-gon

```

1 -- Step 1: Define the ring and parameters
2 kk = QQ;
3 RingP3 = kk[x_0..x_3];
4 d = 2; -- Degree of the hypersurface
5 nrPoints = 4;
6
7 -- Step 2 and 3: Generate random points and create pairs of
  subsequent points
8 -- Function for generating i random points in P^3
9 randomPoints = i -> (
10   v := for j in 0..3 list random(kk);
11   -- Ensure the point is not zero
12   while all(v, x -> x == 0) do (
13     v = for j in 0..3 list random(kk)
14   );
15   v
16 )
17 -- Pick nrPoints random points in P^3
18 myPoints = apply(nrPoints, randomPoints);

```

```

19 -- Create pairs of subsequent points, and also the pair consisting
    of the last and first point
20 subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i
    , myPoints#(i+1)};
21 -- Add final pair
22 subsequentPairs = append(subsequentPairs, {myPoints#0, myPoints#(
    myPoints-1)});
23
24 -- Step 3: Generate matrices from point pairs
25 -- Function to convert two points to a matrix with first row {x_0
    .. x_3}
26 pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points
    #1};
27 -- Convert each pair of points to a matrix
28 pointsMatrix = apply(subsequentPairs, pointsToMatrix);
29
30 -- Step 4: Get the ideals of the lines
31 -- Given a matrix with first row {x_0 .. x_3} and next two rows
    points, return the ideal of the line through the points
32 lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
33 allLines = apply(pointsMatrix, lineFromPoints);
34
35 -- Step 5: Create the ideal of the union of all lines
36 -- Intersect the ideals of the lines to get ideal of the union of
    all lines
37 Iz = intersect allLines;
38
39 -- Step 6: Pick a degree d element in Iz, which defines a surface
    containing all the lines, and check properties
40 -- Pick a random degree d element in Iz
41 f = random(d, Iz);
42 -- Create the variety defined by f
43 Vf = variety ideal f;
44
45 -- Check if the surface is smooth and not the zero polynomial
46 codimension = codim singularLocus ideal f;
47 isZero = (f == 0)
48 isSingular = (codimension < 4)
49 isSmooth(Vf) and not isZero

```

We'll go through the code step by step.

Step 1: We begin by defining the field and the polynomial ring in four variables over this field. We then define the degree d of the surfaces we want to consider, and the number of points to use, which will be the same as the number of lines in the k -gon.

Step 2: We define a function to pick a given amount of random points in our quotient ring, ensuring they are not zero. We then apply the function to get the points, and gather the points into pairs of subsequent points so that we can later define lines between these.

Step 3: Next, we define a function for turning two points into a matrix. The function takes two points and returns a matrix, as follows:

$$((a, b, c, d), (d, e, f, g)) \mapsto \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ a & b & c & d \\ d & e & f & g \end{pmatrix}$$

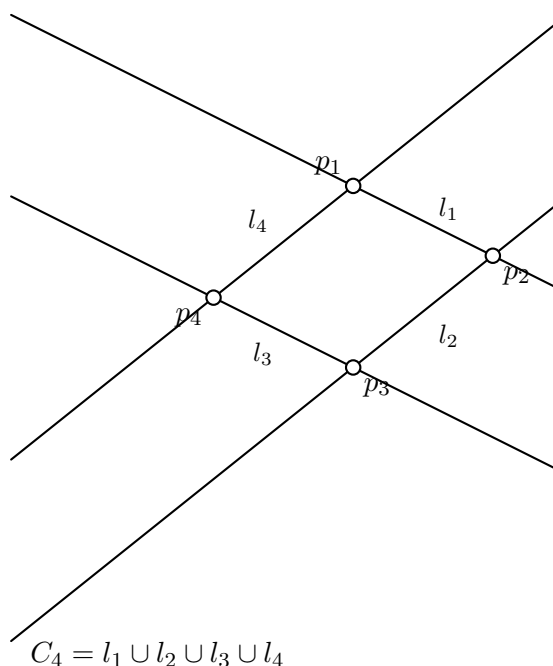
Figure 2.3: A 4-gon in $\mathbb{P}^3$, drawn on a plane.

fig:4-gon

Step 4: Now we can define the ideal of the line between the two points by taking the 3×3 minors of this matrix.

Step 5: Next we take the intersection of all the line ideals, to get the ideal of $Z = \bigcup l_i$,

Step 6: We choose a random member of $I_Z(d)$ and compute if this defines an empty, singular, or smooth surface.

Results: We have been interested in finding the maximum number of lines k such that we can find a smooth surface of degree d including a k -gon. The results are summarized in the following table:

Improve code explanation, and include results. Focus on main ideas of the code?

2.5 Cohomology of a 4-gon

:cohomology

As mentioned earlier, to get an exact value for $h^0(\mathbb{P}^3, \mathcal{I}_{C_k})$, we want to compute $h^1(\mathbb{P}^3, \mathcal{I}_{C_k})$. We will now explore this issue for the case of a 4-gon and looking at a suitable resolution of the ideal sheaf of the 4-gon. First we claim that the intersection points of a 4-gon can, by an automorphism of $\mathbb{P}^3$, be expressed as the points $(1 : 0 : 0 : 0)$, $(0 : 1 : 0 : 0)$, $(0 : 0 : 1 : 0)$, $(0 : 0 : 0 : 1)$. Then the 4-gon can be expressed as the union of the lines $l_1 = V(x_2, x_3)$, $l_2 = V(x_0, x_3)$, $l_3 = V(x_0, x_1)$, $l_4 = V(x_1, x_2)$. Now the ideal of the 4-gon is given by $I_{C_4} = (x_2, x_3) \cap (x_0, x_3) \cap (x_0, x_1) \cap (x_1, x_2) = (x_1x_3, x_0x_2)$. We justify our choice of points in the following lemma.

Lemma 2.5.1. *Let C_4 be a 4-gon in $\mathbb{P}^3$. Then there exists an automorphism of $\mathbb{P}^3$ such that the intersection points of C_4 are given by the points $(1 : 0 : 0 : 0)$, $(0 : 1 : 0 : 0)$, $(0 : 0 : 1 : 0)$, $(0 : 0 : 0 : 1)$.*

Proof. Let p_1, p_2, p_3, p_4 be the intersection points of C_4 . To be in general position in $\mathbb{P}^3$, we require that no three of the points are collinear, and that the points do not all lie in a plane. We can then find a projective transformation sending p_1 to $(1 : 0 : 0 : 0)$, p_2 to

$(0 : 1 : 0 : 0)$, and p_3 to $(0 : 0 : 1 : 0)$. Since no three of the points are collinear, p_4 cannot be sent to a point on the line between any two of the other points. Furthermore, since the points do not all lie in a plane, p_4 cannot be sent to a point in the plane spanned by any three of the other points. Then we can send p_4 to $(0 : 0 : 0 : 1)$, and we are finished. $\square$

We will now look at a resolution of the ideal sheaf of the 4-gon. From the discussion above, we have that $I_{C_4} = (x_1x_3, x_0x_2)$. This ideal has the following free resolution:

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^3}(-4) \xrightarrow{\begin{pmatrix} -x_0x_2 \\ x_1x_3 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4} \rightarrow 0.$$

Finish argument.

2.6 Surfaces including k-chains

Definition 2.6.1. A k -chain is a union of k lines $l_1, l_2, \dots, l_k$ such that l_i intersects l_{i+1} for $1 \leq i < k$. We will write $C_k = \bigcup_{i=1}^k l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \leq i < k$.

Notice that a k -chain is a $(k+1)$ -gon excluding the line l_k intersecting l_1 . One can then modify the code above to create k -chains instead of $(k+1)$ -gons, by excluding the final pair of points. The results from this are similar to those of k -gons, in that for higher degrees, we find smooth surfaces, up until we find no surfaces. There are however interesting exceptions for degree 2 and 3 surfaces. Here we get singular surfaces for 4-chains and 7-chains. We will handle the cases separately.

Include results for k -chains

Proposition 2.6.2. For a general 4-chain in $\mathbb{P}^3$, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.

Proof. First, assume that S_2 is smooth. Since S_2 is a smooth quadric surface in $\mathbb{P}^3$, it is of the form $S_2 = \mathbb{P}^1 \times \mathbb{P}^1$. This means that S_2 has two rulings, meaning that there are two families of lines on S_2 , such that any two lines in the same family do not intersect, and any two lines in different families intersect at exactly one point. Denote the two families as F_1 and F_2 . Then we can see that a 4-chain must consist of two lines from F_1 and two lines from F_2 , since any two lines in the same family do not intersect. This means that the 4-chain must be of the form l_1, l_2, l_3, l_4 where $l_1, l_3 \in F_1$ and $l_2, l_4 \in F_2$. However, this means that l_4 intersects l_1 , since they are in different families, meaning that the 4-chain is in fact a 4-gon. This is a contradiction, since this is not a general 4-chain. Thus, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain. $\square$

Proposition 2.6.3. For a general 4-chain in $\mathbb{P}^3$, we can find a singular quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.

Proof. Find a nice example of a singular quadric surface including a general 4-chain. May be able to use example from m2. It takes the five points $(1 : 0 : 0 : 0), (0 : 1 : 0 : 0), (0 : 0 : 1 : 0), (0 : 0 : 0 : 1), (1 : 1 : 1 : 1)$, which are the intersection points of a 4-chain, and gives the singular quadric surface $V(xw - yw)$ including the 4-chain. $\square$

Cite
Beauville

Definition 2.6.4. We define $Q_m \subseteq \mathbb{G}(1, 3)^m$ to be the subset of m -tuples of lines $(l_1, \dots, l_m)$ such that they form an m -chain.

lem:dimQm

Lemma 2.6.5. Q_m is an irreducible variety of dimension $3m + 1$ for $m \geq 1$ (Do we have to require $m \geq 1$, or is it obvious!?).

Irreducibility is discussed in EH 3.3.1, saying Schubert cells and their closures are irreducible.

Proof. We observe that Q_m can be constructed as a product of Grassmannians, with a condition of intersection added for $m \geq 2$. That is, if we are given a line $L \in \mathbb{G}(1, 3)$, we can consider the space of lines intersecting L , denoted as $\Sigma_L = \{l \in \mathbb{G}(1, 3) \mid l \cap L \neq \emptyset\}$. Then $Q_m = \mathbb{G}(1, 3) \times \Sigma_L^{m-1}$. For $m = 1$, we have that $Q_1 = \mathbb{G}(1, 3)$, which is irreducible of dimension 4. Now, assume that the statement holds for $m - 1$. Consider the projection map $\pi : Q_m \rightarrow Q_{m-1}$ given by $\pi(l_1, \dots, l_m) = (l_1, \dots, l_{m-1})$. The fiber over a point is given by $\pi^{-1}(l_1, \dots, l_{m-1}) = \{l \in \mathbb{G}(1, 3) \mid l \cap l_{m-1} \neq \emptyset\}$.

Should we denote the fiber over a point as the whole m -tuple, or just the last line? I think the first is more correct.

Note that for a given line L , Σ_L can be constructed as the image of the incidence correspondence $\Gamma = \{(l, p) \in \mathbb{G}(1, 3) \times L \mid p \in l\}$ to $\mathbb{G}(1, 3)$. The fiber of the projection map $f : \Gamma \rightarrow L$ is given by the set of lines through a point $p \in L$, which is isomorphic to $\mathbb{P}^2$. Then the fiber of each point in L is irreducible of dimension 2, and since f is proper, we have that Γ is irreducible of dimension 3, meaning that Σ_L is irreducible of dimension 3. Thus, Q_m is the product of $m - 1$ irreducible varieties of dimension 3, and one Grassmannian of dimension 4, so Q_m is irreducible of dimension $4 + 3(m - 1) = 3m + 1$. This completes the proof. $\square$

Update proof with notes from 09.10. Don't think this is rigorous enough to show irreducibility.

Now we will look at the case of 7-chains in cubic surfaces.

Proposition 2.6.6. For a general 7-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain.

Proof. Let $\mathcal{M}$ be the space of smooth cubic surfaces in $\mathbb{P}^3$. This is an open subset of $\mathbb{P}(\mathbb{C}[x_0, \dots, x_3]_3) \cong \mathbb{P}^{19}$. Since $\mathcal{M}$ is an open subset, it shares the dimension, i.e. $\dim \mathcal{M} = 19$. Let $\mathcal{N} = \{(S_d, l_1, \dots, l_7) \in \mathcal{M} \times \mathbb{G}(1, 3)^7 \mid \mathcal{L} \subset S_d, \mathcal{L} \text{ is a } k\text{-chain}\}$ be the incidence variety of smooth cubic surfaces including a general 7-chain. We have the following diagram:

Why open?

$$\begin{array}{ccc} & \mathcal{N} & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ \mathcal{M} & & \mathbb{G}(1, 3)^7 \end{array} \quad (2.1)$$

Should we use $\mathcal{Q}_7$ instead of $\mathbb{G}(1, 3)^7$?

The projection map $\pi_1 : \mathcal{N} \rightarrow \mathcal{M}$ is given by $\pi_1(S_d, l_1, \dots, l_7) = S_d$. Then the fiber over a point $S_d \in \mathcal{M}$ is given by the set of 7-chains included in S_d . Since a smooth cubic surface includes exactly 27 lines, we can find a 7-chain by choosing 7 lines from these 27. We know from Proposition 2.6.8(!?) that it is possible to find a 7-chain, and so this fiber is not empty. Furthermore, there are only a finite amount of 7-chains when there is a finite amount of lines. Then the dimension of the fiber

Refer to result

Result in this section

$\dim \pi_1^{-1}(S_d)$ is 0, being a finite set of 7-chains. Thus, from Theorem 0.1.3, we have that $\dim \mathcal{N} = \dim \mathcal{M} + \dim \pi_1^{-1}(S_d) = 19 + 0 = 19$.

Refer to theorem about dimension of fiber.

Now we are interested in the fiber of the set of 7-chains in $\mathbb{P}^3$, $\mathcal{Q}_7 \subseteq \mathbb{G}(1, 3)^7$. By Lemma 2.6.5, we have that $\dim \mathcal{Q}_7 = 22$, and so the projection map $\pi_2 : \mathcal{N} \rightarrow \mathbb{G}(1, 3)^7$ given by $\pi_2(S_d, l_1, \dots, l_7) = (l_1, \dots, l_7)$ cannot be surjective on (dominate!?) $\mathcal{Q}_7$. This means that taking a general 7-chain $C_7 \in \mathcal{Q}_7$, and considering the fiber $\pi_2^{-1}(C_7)$, we do not expect this to be non-empty, meaning we will not have a surface including it. Thus, for a general 7-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain. $\square$

Proof should include: projection is surjective (every smooth cubic surface has at least a 7-chain, shown in this section), implies $\dim N \geq \dim M$. Then fiber is finite, so $\dim N \leq \dim M$. Prop 10.26?

Again, as in the case for 4-chains in quadric surfaces, computations show that we can find a singular cubic surface including a general 7-chain.

Proposition 2.6.7. *For a general 7-chain in $\mathbb{P}^3$, we can find a singular cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain. (!?)*

Proof. Find a nice example of a singular cubic surface including a general 7-chain. $\square$

Need to rework entire section above, should be 6-chains, not 7-chains.

Note that if we instead asked whether a smooth cubic surface in $\mathbb{P}^3$ can include a 7-chain, the situation is different. In this case, we can indeed find a 7-chain such that it is included in the surface. In fact, we know we could find an 11-chain in this case, and a 12-gon. We show this in the following proposition.

Maybe move this proposition earlier, to use in proof. Maybe lemma? Also interesting in itself, so maybe not.

prop:12-gon

Proposition 2.6.8. *For a smooth cubic surface in $\mathbb{P}^3$, we can find a 12-gon such that it is included in the surface.*

Proof. Recall from Proposition 1.9.2 that a smooth cubic surface in $\mathbb{P}^3$ is isomorphic to the blow-up of $\mathbb{P}^2$ in 6 points. It may be useful to again consider the picture in figure Fig. 1.1 on page (!?). The surface includes 27 lines, which are precisely the exceptional divisors of the blow-up, the strict transforms of the 15 lines through pairs of points, and the strict transforms of the 6 conics through 5 of the points (Proposition 1.9.3). We can then find a 12-gon by considering the following lines: Let $E_1, E_2, E_3, E_4, E_5, E_6$ be the exceptional divisors, let L_{ij} be the strict transform of the line through points p_i and p_j , and let C_i be the strict transform of the conic through all points except p_i . Then we can consider the 12-gon given by the lines $E_1, L_{12}, E_2, L_{23}, E_3, L_{34}, E_4, L_{45}, E_5, L_{56}, E_6, C_1, L_{16}$. This is indeed a (general!?) 12-gon since each line intersects exactly two other lines in the set. $\square$

By removing one of the lines from the 12-gon, we immediately get the following:

Corollary 2.6.9. *For a smooth cubic surface in $\mathbb{P}^3$, we can find an 11-chain such that it is included in the surface.*

This implies that the k -chains longer than 7 included in the smooth cubic surfaces are somehow special.

Fill out later.

Fix $G(k, n)$. k is overloaded. Maybe change to using m -chains and m -gons?

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